

Mohammad Abu Thaher Chowdhury

📍 Troy, NY | 📞 956-576-2997 | ✉️ chowdm4@rpi.edu | 🎓 Google Scholar | 🌐 Thaher | 🌐 Thaher

Highlights

- 4+ years of research experience in cosmological simulations, galaxy formation, and fuzzy/scalar-field dark matter using GPU-accelerated high-performance computing (HPC).
- Developed a **MATLAB GPU code** for baryonic effects on fuzzy dark matter; currently extending to a **distributed GPU-parallel workflow** for large-scale 3D simulations.
- Designed **gravitational-wave data analysis pipelines** with Particle Swarm Optimization (PSO), adaptive spline fitting, and wavelet-based techniques; improved glitch subtraction for LIGO/Virgo data.
- Strong background in astrophysics, gravitational waves, quantum computing, and big-data driven simulation analysis.
- Technical skills in **Python, MATLAB, LaTeX, HDF5, Git**; proficient with **GAMER, GIZMO**, and GPU/cluster workflows.
- 2 peer-reviewed publications, including *Classical and Quantum Gravity* (2023) and a review on gravitational-wave glitch subtraction (arXiv, 2024).
- Awards: **Presidential Graduate Research Fellowship** (UTRGV, 2021–2022) and **Bangladesh Ministry of Education Fellowship** (2014).
- Leadership: **Qiskit Advocate** (IBM), General Secretary of Astronomy Club (2022), Organizer/Trainer at SUST Career Club.

Experience

Galaxy and Star Formation Lab, RPI

Troy, NY

Graduate Research Assistant

Jan 2023 – Present

- Conduct cosmological simulations (GAMER, GIZMO) to study galaxy formation in scalar/fuzzy dark matter frameworks.
- Analyze soliton core–halo structure and baryonic effects using GPU-accelerated high-performance computing.
- **Developed a MATLAB GPU code** for baryonic effects on fuzzy dark matter; extending to a **distributed GPU-parallel workflow** for large 3D simulations.
- Developed **PHANTOM**, an open-source MATLAB toolbox for halo-profile analysis, concentration modeling, and reproducible dark matter simulation workflows.

Department of Physics, Applied Physics, and Astronomy, RPI

Troy, NY

Graduate Teaching Assistant

Jan 2023 – Present

- Assisted in undergraduate courses (Mechanics, Modern Physics, Advanced Quantum Mechanics); graded assignments and lab reports.
- Supported graduate-level courses in Quantum Mechanics I and Statistical Mechanics; conducted tutoring sessions.

Gravitational Wave Group, UTRGV

Edinburg, TX

Graduate Research Assistant

Jan 2021 – Dec 2022

- Researched gravitational-wave data conditioning and glitch subtraction for LIGO/Virgo.
- **Built analysis pipelines using PSO** for parameter estimation and denoising.
- Developed adaptive spline and wavelet techniques for glitch mitigation; presented results at APS and TACCSTER.

Astronomy & Physics Labs, UTRGV

Edinburg, TX

Teaching & Lab Assistant

Jan 2022 – Dec 2022

- Led astronomy/physics labs, supervised student work, and guided experiments in mechanics, electricity, magnetism, and optics.

- Performed stellar data reduction and analysis; built a MATLAB tool for measuring speed of light.

Department of Mathematics and Natural Sciences, BRAC University

Dhaka, Bangladesh

Lecturer

Jan 2018 – Dec 2020

- Taught undergraduate courses and labs in Mechanics, Electricity & Magnetism, and Optics; supervised projects.

Department of Physics, Universal College Sylhet

Sylhet, Bangladesh

Lecturer

Aug 2016 – Dec 2017

- Taught physics courses and labs; served on the admissions committee.

PP & PDC Research Lab, BCSIR

Dhaka, Bangladesh

Research Assistant

Jul 2015 – Jul 2016

- Prepared samples and analyzed data using XRD and SEM.

Technical Skills

Programming & Tools: Python, MATLAB, Octave, PostgreSQL, LaTeX, Git, HDF5, Microsoft Office

Data Science & Analysis: NumPy, SciPy, Matplotlib, data analysis, curve fitting, statistical modeling, scientific visualization

Databases: PostgreSQL, structured data querying, relational database fundamentals

HPC & Scientific Computing: GPU-accelerated computing, job arrays, profiling, distributed workflows (in progress), large-scale simulation analysis

Simulations: GAMER, GIZMO, jaxion

Quantum Computing: Qiskit, PennyLane

Publications (Selected)

- M. A. T. Chowdhury, Victor Robles *Fuzzy Dark Matter Halo Structure from Filamentary Collapse to Virialization: Core-Halo Mass Relation and Dwarf Galaxy Kinematics*. *In preparation*, 2026.
- Chowdhury, M.A.T. (2026). *PHANTOM: A MATLAB and Octave Toolbox Connecting Linear Field Statistics to Dark Matter Halo Observables* Under review, 2026
- Chowdhury, M.A.T. (2026). *PHANTOM: Profile and Halo Analysis for Numerous Theoretical Dark Matter Observables* [Software]. Zenodo. doi:10.5281/zenodo.19700919
- Chowdhury, M.A.T., Mohanty, S.D. (2023). *Glitch subtraction from gravitational wave data using adaptive spline fitting*. *Classical and Quantum Gravity*, 40, 125001.
- Chowdhury, M.A.T. (2024). *Advancements in glitch subtraction systems for enhancing gravitational wave data analysis: A brief review*. arXiv:2406.01318.
- For full publications and talks, please visit my [google scholar page](#).

Selected Projects

PHANTOM: Profile and Halo Analysis for Numerous Theoretical Dark Matter Observables GitHub / 2025–Present

- Developed an open-source MATLAB toolbox and Octave package for computing dark matter halo structural properties across multiple theoretical models.
- Implemented halo concentration, density-profile, and core-halo mass relation modules for fuzzy dark matter and related cosmological analyses.
- Built the package with documentation, testing workflows, citation metadata, and reproducible research support for public release.

GPU-Accelerated FDM Baryonic-Effects Code (MATLAB)

RPI

- Implemented GPU kernels to accelerate soliton-halo analyses; refactored I/O to HDF5; modularized post-processing for density and velocity diagnostics.

- Currently extending to distributed/GPU-parallel execution across nodes for large 3D grids.

Gravitational-Wave Data Analysis with PSO

UTRGV

- Designed PSO-based routines integrated with adaptive spline and wavelet shrinkage for glitch mitigation.
- Benchmarked convergence vs. baseline methods; improved detection fidelity on contaminated data.

Education

Rensselaer Polytechnic Institute (RPI), Troy, NY

Jan 2023 – Exp. Dec 2027

Ph.D. in Physics, GPA: 4.00/4.00

Courses: Quantum Mechanics II, Electrodynamics, Cosmology, Astrophysics of Galaxies, Interstellar Medium, etc.

University of Texas Rio Grande Valley (UTRGV), Edinburg, TX

Jan 2021 - Dec 2022

M.S. in Physics, GPA: 3.82/4.00

Courses: Classical Mechanics, Statistical Mechanics, Gravitational Wave Data Analysis, Optics, etc.

Shahjalal University of Science and Technology (SUST), Sylhet, Bangladesh

Aug 2009 – Jun 2015

B.Sc. and M.Sc. in Physics, GPA: 3.28/4.00 & 3.86/4.00

Awards & Leadership

Awards: Presidential Graduate Research Fellowship, UTRGV (2021–2022); Ministry of Education Fellowship, Bangladesh (2014)

Leadership: Qiskit Advocate, IBM (2025–Present); General Secretary, Astronomy Club (2022); Organizer & Trainer, SUST Career Club (2013–2015)